

RGB-D



Frontend



Qwen

"Table, Window,
Lamp, Wall,
Stool, ..."

Grounded-
SAM



Async - Refinement

Pair Scheduler

Frame Scheduler

Critic Agent



Candidates: Obj1, Obj2
Reason: "both objects have a high IoU and look visual, like fragments of the same carpet"
Decision: Merge

Description Agent



Description: "round table",
Material: "wood"
Color: "medium brown"
State: "upright"
Texture: "smooth"

Incremental 3D Scene Graph

Merging
Nodes

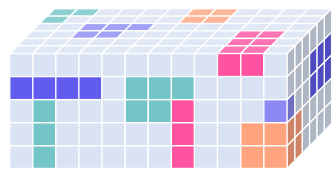
Refining
Nodes

Adding
Nodes

Adding
Relations

Backend

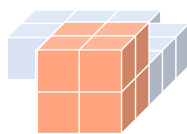
Filtering & Projection



2D to 3D

Association

Geometric Score



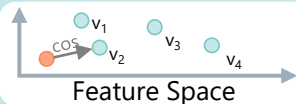
Feature Score



Label Voting



Feature Selection



Relation Generation



Heuristics

